

1 Introduction & Substrate Specifications

A microstrip Wilkinson Power Divider was designed in Keysight ADS for center frequency $f_0 = 5.8$ GHz. The design uses an FR-4 substrate (MSub1) and delivers equal power division (3 dB split) with high output port isolation.

Table 1: Substrate (MSub1) and Transmission Line Synthesis Parameters

Substrate Parameter	Value	Microstrip Section	Dimensions ($W \times L$)
Substrate Height (H)	1.6 mm	50 Ω Input Line (TL1)	3.209660 mm $\times$ 7.094260 mm
Relative Permittivity (ϵ_r)	4.3	$\sqrt{2}Z_0 = 70.71 \Omega$ Quarter-Wave (TL2, TL3)	1.690890 mm $\times$ 7.318920 mm
Conductor Thickness (T) / Cond.	0.035 mm / 5.8×10^7 S/m	Isolation Resistor (R_1)	100.0 Ω
Loss Tangent ($\tan \delta$) / Roughness	0.025 / 0.0001 mm	Port Terminations (Z_0)	50.0 Ω (TermG1, TermG2, TermG3)

2 Design Synthesis & Circuit Equations

The power divider splits input power equally at Port 1 into Ports 2 and 3 using $\lambda/4$ transmission lines ($Z = \sqrt{2}Z_0 \approx 70.7 \Omega$) and a 100 Ω isolation resistor:

$$Z_{\lambda/4} = \sqrt{2} \times 50 \Omega = 70.71 \Omega, \quad R_{\text{iso}} = 2 \times 50 \Omega = 100 \Omega, \quad L_{\lambda/4} \approx \frac{c}{4f_0\sqrt{\epsilon_{\text{eff}}}} = 7.319 \text{ mm} \quad (1)$$

3 ADS Schematic Topology

The schematic setup in Keysight ADS covers the frequency sweep from 5.0 GHz to 6.5 GHz with a step size of 0.001 GHz (SP1).

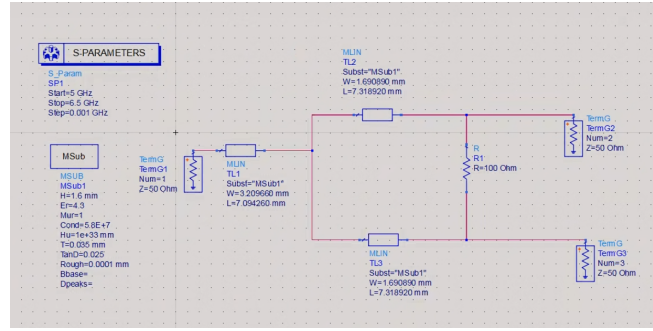


Figure 1: Keysight ADS schematic capture of the 5.8 GHz Wilkinson Power Divider.

4 ADS Simulation Results

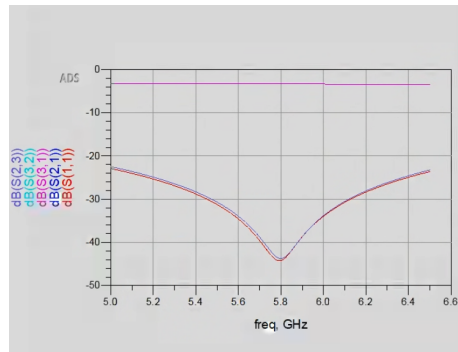


Figure 2: Simulated S-parameters showing Return Loss (S_{11}), Insertion Loss (S_{21}, S_{31}), and Isolation (S_{23}, S_{32}).

4.1 Performance Summary at 5.8 GHz

Table 2: ADS S-Parameter Simulation Performance at Center Frequency $f_0 = 5.8$ GHz

Parameter	Insertion Loss (S_{21}, S_{31})	Input Return Loss (S_{11})	Isolation (S_{23}, S_{32})	Fractional Bandwidth (< -20 dB)
ADS Simulation	-3.05 dB (3 dB split + loss)	-44.2 dB	-43.8 dB	5.0 – 6.5 GHz ($> 25\%$)